

PROFESSIONAL SUMMARY

Research Scientist at Ivo working on supervised fine-tuning, prompt optimization, and post-training of LLMs for knowledge-intensive domains such as law. Previously 6+ years at Lawrence Livermore National Laboratory, adapting foundation models and graph neural networks to high-stakes scientific domains. Published at NeurIPS and ICLR. Chess Grandmaster.

SKILLS AND COMPETENCIES

Fine-tuning & PEFT | Multimodal Foundation Models | LLMs | Graph Neural Networks | Uncertainty Quantification | Scientific ML
PyTorch | PyTorch Geometric | Hugging Face | Pandas | scikit-learn | Databricks | BioPython | PyTorch Lightning | Slurm

EXPERIENCE

Research Scientist, Ivo 2026 - present

- Developing supervised fine-tuning, prompt optimization, and post-training for LLMs in knowledge-intensive domains such as law.

Senior Research Scientist, Lawrence Livermore National Laboratory 2020 - 2026

Senior Research Scientist 2022 - 2026

- Developed new machine learning algorithms for making trustworthy predictions leveraging scientific knowledge.
- Led applied machine learning initiatives fine-tuning multimodal foundation models and leveraging them together with graph neural networks on cross-functional teams in bioinformatics, software analysis, and power systems.
- Shaped direction of DOE-funded project on inferring temporal knowledge graphs using LLMs as co-principal investigator.
- Mentored 10 graduate students and built long-term collaborations with academic institutions that resulted in 13 publications.

Postdoctoral Researcher 2020 - 2022

- Defined new GNN research direction and presented tutorials on graph representation learning of non-local structural similarity.

Data Science Research Intern, Adobe Research Winter 2019

- Improved user stitching accuracy by 5% and space-efficiency by 50x for millions of users using compressed graph embeddings.

Research Intern, Information Sciences Institute Summer 2019

- Used social media network structure and content to improve cyberbullying detection accuracy by up to 43%.

SELECTED RESEARCH WORKS

Safe and Robust Graph Machine Learning

- Improved performance on safety-critical graph ML tasks with anchoring methods for GNNs (ICLR 2024) and conformal prediction (SDM 2025) for uncertainty quantification. Reduced calibration error for graph classification under distribution shift by up to 30%.
- Defined new GNN research direction of learning on heterophilous graphs (NeurIPS 2020; Data Engr. Bulletin 2023) and characterized performance discrepancies based on heterophily (LoG 2024).
- Developed and theoretically justified new algorithm for up to 80% more robust graph matching (SDM 2021) and a 10x faster multiscale framework (CIKM 2022).

Knowledge-Guided Machine Learning

- Showed theoretical and empirical benefit of domain-specific data augmentation for graph self-supervised learning (NeurIPS 2022).
- Developed new methods using multiscale topological priors for segmentation of scientific images (TVCG 2023; ICIP 2023).
- Proposed learnable graph-structured knowledge for few-shot image classification (WACV 2023), achieving up to 10% higher accuracy than the state of the art in few-shot generalization to new datasets.

EDUCATION

PhD Computer Science, University of Michigan 2015 - 2020

AB Economics & Mathematics; MS Computer Science, Washington University in St. Louis 2011 - 2015

SELECTED HONORS

- Best Student Paper: ICDM 2019 ("Distribution of Node Embeddings as Multiresolution Features for Graphs")
- Best Program Committee Member for outstanding reviewing service: WebConf 2021, 2024

ADDITIONAL ACTIVITIES

Grandmaster, Professional Chess

- Earned grandmaster title, the highest title in the game, in 2024 while working full-time in AI.
- Coached multiple advanced students, one of whom achieved the national master title in 2025.
- Spoke to business leaders at Ridge Ventures about insights from rising to elite level in chess alongside demanding parallel pursuits.

Chess Rating System Audit

- Demonstrated demographic-specific bias in the chess rating system across >10M games: up to 20% underestimation of upset probabilities. Highlighted practices in the chess world contributing to bias.